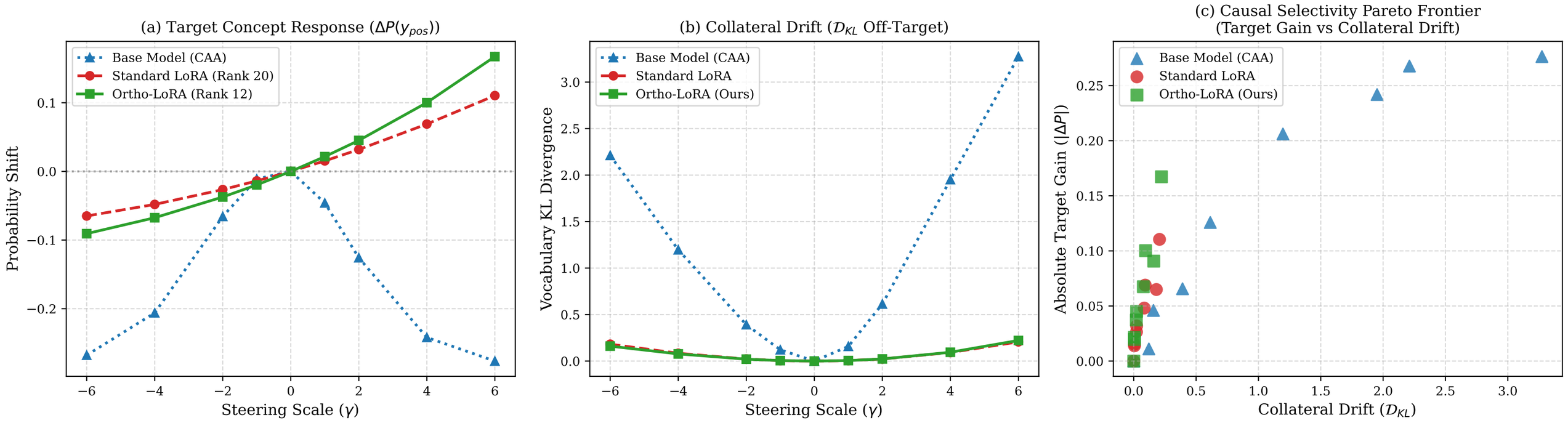
(a) Target Concept Response ($\Delta P(y_{pos})$)
Base Model (CAA)
Standard LoRA (Rank 20)
Ortho-LoRA (Rank 12)
Probability Shift
Steering Scale (γ)
(b) Collateral Drift ($\mathcal{D}_{KL}$ Off-Target)
Base Model (CAA)
Standard LoRA
Ortho-LoRA (Ours)
Vocabulary KL Divergence
Steering Scale (γ)
(c) Causal Selectivity Pareto Frontier
(Target Gain vs Collateral Drift)
Base Model (CAA)
Standard LoRA
Ortho-LoRA (Ours)
Absolute Target Gain ($|\Delta P|$)
Collateral Drift ($\mathcal{D}_{KL}$)